

6G-PFIP: Privacy-Preserving and Personalized Federated Intelligent Paging for 6G Wireless Networks

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Abstract—Paging is anticipated to play a pivotal role in realizing the 6G vision for the IoE (Internet of Everything) and uRLLC (ultra-reliable low-latency communications) by enhancing service quality and improving resource efficiency. Nevertheless, as 6G architectures evolve toward large-scale, distributed architectures, existing rule-driven, centralized paging mechanisms face severe challenges, including inefficiencies due to massive mobile-device access, incompatibility with distributed network architectures, and significant privacy concerns. To address these challenges, this paper proposes a privacy-preserving and personalized federated intelligent paging mechanism for 6G Networks (6G-PFIP). 6G-PFIP introduces a novel intelligent decision model to enhance paging efficiency for large-scale user equipment (UE). A personalized hierarchical federated paging architecture (HFPA) is further constructed to improve architectural compatibility and privacy by decomposing the centralized paging process into distributed layers. HFPA effectively addresses the challenge of Non-IID user data in 6G by designing region-level and local-level personalization and fairness-aware aggregation strategies. Furthermore, HFPA integrates local differential privacy (LDP) on demand to perturb shared model parameters, thereby mitigating the risk of privacy leakage. Experiments on a 6G Telecom Dataset show that 6G-PFIP achieves significant signaling resource savings (71.33%) and delay reductions (34.78%), while rigorous analysis confirms its privacy guarantees.

Index Terms—6G Networks, Intelligent Paging, Efficiency, Privacy-Preserving, Federated Learning.

I. INTRODUCTION

THE sixth-generation mobile network (6G) envisions the IoE and uRLLC as its pivotal objectives, aiming to support massive device connections in high-mobility scenarios [1]. Against this overarching vision, the paging mechanism plays a critical role in meeting increasingly stringent requirements for scalability, latency, energy efficiency, and privacy [2].

Paging serves as a fundamental control mechanism in wireless communication systems [3]. The existing 5GC paging procedure defined by 3GPP is illustrated in Fig. 1 and can be summarized in four phases:

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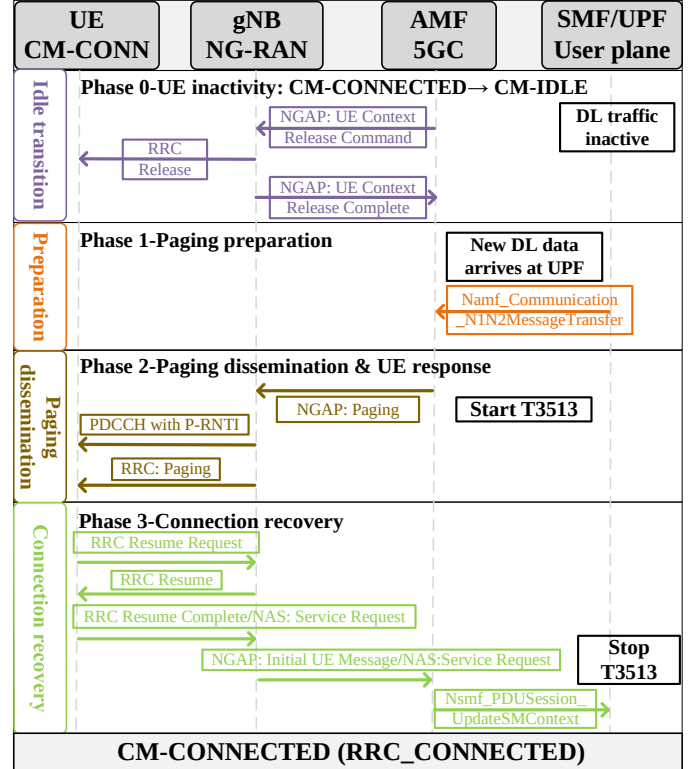


Fig. 1. Schematic of the 3GPP-Defined the existing Paging Procedure.

❶ **Phase 0: Idle Transition (CM-CONNECTED → CM-IDLE).** When the downlink traffic of a UE becomes inactive, the Access and Mobility Management Function (AMF) sends an NGAP: UE Context Release Command to the next-generation NodeB (gNB), which then issues an RRC Release to free radio resources and replies with an NGAP: UE Context Release Complete. The UE transitions to the CM-IDLE state, where the core network retains only coarse-grained Tracking Area information.

❷ **Phase 1: Paging Preparation.** When new downlink data arrives at the UPF for an idle UE, the SMF notifies the AMF via Namf_Communication_N1N2MessageTransfer, triggering the AMF to initiate the paging procedure based on the UE's registered Tracking Area (TA) information.

❸ **Phase 2: Paging Dissemination and UE Response.** The AMF sends NGAP: Paging messages to the gNBs within the UE's registered Tracking Area List (TAL) and starts the T3513 timer. The gNB transmits paging indications via the PDCCH, scrambled with the P-RNTI. Upon detection during

its discontinuous reception (DRX) cycle, the UE receives the RRC: Paging message and identifies itself as the intended recipient.

④ Phase 3: Connection Recovery. The UE initiates the connection recovery by sending an RRC Resume Request and receives RRC Resume from the gNB. It then completes the procedure via RRC Resume Complete, which carries the NAS Service Request. The gNB forwards the NAS Service Request to the AMF via NGAP: Initial UE Message. Upon reception, the AMF stops the T3513 timer and invokes Nsmf_PDUSession_UpdateSMContext to resume the PDU session. The UE then transitions back to the CM-CONNECTED (RRC_CONNECTED) state [4].

However, as a typical rule-driven process, the existing paging mechanism faces several fundamental limitations [5]. Specifically, current 3GPP paging relies on centralized control by the AMF [6]. With the rapid rise of 6G devices and increasingly complex user mobility [7], rule-based mechanisms fail to track dynamic spatio-temporal UE behaviors [8]. Consequently, networks often resort to coarse-grained paging over entire TAs or TALs, resulting in excess signaling overhead and higher access latency [9]. Moreover, the shift to distributed 6G architectures [10] further reveals the limits of centralized paging. These mechanisms struggle with decentralized control and large-scale coordination. In such environments, regionally collected user data exhibits significant statistical heterogeneity and becomes non-independent and identically distributed (Non-IID), meaning data samples vary greatly across regions [11], complicating accurate and adaptive paging. Additionally, existing paging procedures reveal sensitive location information, such as serving gNBs and TA/TAL identifiers, in plaintext during signaling. This results in privacy risks, which intensify under service-oriented 6G architectures with greater system openness and interface exposure. Overall, the evolution of paging mechanisms faces **three major challenges**.

CHALLENGE 1. Paging inefficiency, as a key challenge, has garnered increasing attention, motivating efforts toward more effective optimization strategies [12]. The typical approach involves collecting raw data, such as UE trajectories and channel quality reports, on the network side to support centralized decision-making. For example, Lin et al. [5] construct a beam-based paging list using UE beam measurements and historical paging results. They apply a Markov chain and 0 – 1 programming to identify optimal beam sets, reducing paging latency. While effective in the centralized architecture of 5G, this method is not well-suited to the distributed architecture envisioned for 6G. It also overlooks network-layer paging optimization, which is essential for end-to-end efficiency.

CHALLENGE 2. The centralized design of existing paging mechanisms makes them poorly compatible with the distributed architecture of 6G networks [13]. Federated learning (FL), with its distributed architecture [14], has shown great potential in addressing architectural limitations of existing paging mechanisms. However, applying conventional FL to paging introduces several issues. First, AMFs or gNBs across regions collect UE trajectory data that differ in scale, feature distribution, and label distribution. This heterogeneity makes single-model FL approaches inadequate for personal-

ized paging. Second, standard aggregation methods, such as weighted averaging, overlook differences in local knowledge, allowing data-rich domains to dominate model updates while marginalizing data-scarce ones [15]. This overlooks the value of rare data in enhancing model generalization and robustness. Such unfairness can degrade paging performance in domains with fewer UEs or uncommon mobility patterns, increasing signaling overhead in large heterogeneous networks.

CHALLENGE 3. Privacy leakage has emerged as another key challenge in the evolution of the paging mechanism, driven by the increasing openness of the next-generation communication networks [16]. Regulations such as the General Data Protection Regulation (GDPR) mandate strict protection of personal data and call for stronger privacy safeguards in next-generation wireless systems [17]. The open and service-oriented nature of the distributed network increases the risk of unauthorized access, exposing more edge interfaces to potential compromise [18]. Malicious insiders can access sensitive user information once a wireless communication edge node is compromised. The existing paging mechanism relies on the continuous upload of sensitive UE data to the core network [19], thereby increasing the risk of privacy leakage. Moreover, it lacks effective privacy-preserving measures against data exposure, especially in the presence of compromised or insider nodes. Despite increasing concerns, current paging research has yet to provide practical solutions to address these privacy risks.

To address the aforementioned challenges, we propose a Privacy-Preserving and Personalized Federated Intelligent Paging Mechanism (6G-PFIP). To the best of our knowledge, 6G-PFIP is the first intelligent paging framework to adopt a distributed architecture and to endow the paging system with optional privacy-preserving capabilities. Our main contributions of this paper are summarized as follows:

- **First**, we design an intelligent paging decision model denoted as RelTime-PRE. By predicting UE mobility and narrowing the paging area, RelTime-PRE enables service-oriented decision-making for the paging and reduces access service latency for a large number of heterogeneous UEs.

- **Second**, 6G-PFIP presents a Personalized Hierarchical Federated Paging Architecture with Fairness (HFPA), comprising AMF, Distributed-NWDAF, and NWDAF. It evolves paging from a centralized framework to a distributed, edge-assisted architecture and integrates region-level and local-level personalization, along with fairness-aware aggregation, to improve compatibility across distributed systems. Meanwhile, HFPA effectively preserves privacy by keeping UE data local on edge AMF nodes during training, while only aggregating model parameters hierarchically.

- **Third**, 6G-PFIP provides effective privacy-preserving capabilities while minimizing performance degradation. HFPA enables lightweight privacy protection and can further enhance such capabilities by integrating local differential privacy. To the best of our knowledge, this represents the first attempt to strike a favorable trade-off between paging performance and privacy preservation.

- The effectiveness of 6G-PFIP is validated using a real-world dataset from Shanghai Telecom. Experimental results

demonstrate that 6G-PFIP achieves a paging signaling resource savings of 71.33% and a paging delay reduction of 34.78%, while enabling strong privacy-preserving capabilities.

The remaining sections are organized as follows. Section II provides an overview of prior research on paging mechanisms and 6G federated learning. Section III provides Models and Design Goals. Section IV presents the optimization constraints. Section V details the proposed 6G-PFIP framework. Section VI analyzes its privacy-preserving properties. Section VII presents experimental validation, and Section VIII evaluates the performance of real-world deployment. Section IX concludes the paper.

II. RELATED WORK

The demand for intelligent, privacy-aware network optimization in 6G drives research into paging mechanisms and FL. Paging studies primarily focus on efficiency and reduced signaling overhead. FL is widely studied as a privacy-preserving approach to distributed learning. These approaches complement each other, making FL a promising approach for intelligent paging, but integration remains largely unexplored. Next, we review research on paging and federated learning in 6G and discuss their potential integration.

A. Paging

Arouk and Ksentini group UEs for group-based paging, significantly reducing downlink signaling overhead [20]. Verma et al. group IoT devices by characteristics, enabling the base station to page multiple UEs with a single message [21]. Lin et al. propose an intelligent directional paging framework that leverages Markov chains and 0-1 optimization to perform beam-based searches based on UE residence probability in mmWave 5G NR systems, thereby reducing paging delay and radio resource usage while ensuring high paging success rates [5]. Hoseinitabatabei et al. introduce a Predictive Mobility Management architecture that delegates location prediction and data compression to the UE side, significantly reducing idle-mode signaling overhead in dense cell deployments [22]. Huang et al. present a user-assisted dynamic RAN Notification Area (UD-RNA) configuration scheme that combines offline multi-objective optimization with online delay-sensitive decisions at the UE. In ultra-dense 5G environments, this scheme adaptively balances paging and RNA update signaling, reducing total signaling for UEs in RRC inactive state [23].

Discussion: Most existing studies focus on improving paging efficiency at the physical or protocol level, while relying on centralized architectures and direct access to user data. Such designs limit scalability and raise potential privacy concerns, and lack support for distributed intelligence. However, 6G paging requires a distributed architecture with built-in privacy preservation, which is currently insufficiently addressed in the research literature.

B. Federated Learning in 6G

Federated learning has been repeatedly emphasized in 6G research. Chamitha et al. emphasize that FL provides a promising framework for securing applications and services in future

6G networks [24]. Afroditi et al. note that traditional centralized artificial intelligence (AI) techniques already struggle to keep up with the vast amounts of data in future 6G networks and to address growing privacy concerns. FL has emerged as a key enabler of Trustworthy AI (TAI), enabling distributed network nodes to participate in AI training without sharing raw data, thereby mitigating the risks associated with centralized data processing [25]. Ammar et al. propose a framework that combines FL with privacy-preserving techniques to improve the security of millimeter-wave (mmWave) beam prediction systems in 6G networks [26]. Kun et al. introduce an adaptive edge-aggregation frequency adjustment mechanism and a contribution-based dynamic model-aggregation weight adjustment mechanism to enhance system training efficiency and model performance [27]. Liu et al. propose HoloFed, which combines multi-band (MB)-RHS and federated learning to address the challenges of limited operating bandwidth and limited practical adaptability [28]. Yao et al. design a Byzantine-robust federated learning method (FedSAC) to address Byzantine attacks in AirComp-based FL [29]. Liu et al. propose centralized deep reinforcement learning (CDRL) and federated deep reinforcement learning (FDRL) frameworks to optimize downlink uRLLC transmission in NR-U and WiFi coexistence systems, thereby enhancing NR-U performance while maintaining WiFi reliability [30].

Discussion: Federated learning has been applied to various 6G scenarios. However, its integration into specific network services remains limited. Directly embedding FL into core network functions, such as paging, enables efficient service delivery while preserving privacy. Existing studies mainly focus on general frameworks or application-layer optimization, overlooking such task-oriented integration. Therefore, FL-based solutions should be tightly coupled with practical network services to meet the stringent requirements of 6G systems in terms of latency, overhead, and privacy.

III. MODELS AND DESIGN GOALS

A. System Model

In the proposed system, the paging mechanism involves five types of entities, namely NWDAF, D-NWDAF, AMF, gNB, and UE, as shown in Fig. 2. These entities are organized into two intelligent modules: the Paging Optimization Intelligent Decision and HFWA, whose functionalities are described below.

1) *NWDAF* N : NWDAF analyzes and processes communication data such as location information and application data. In our system, NWDAF aggregates and distributes the global intelligent paging decision model W to the central core network.

2) *D-NWDAF* S_d : D-NWDAF operates similar functions to NWDAF in the distributed core network. In our system, D-NWDAF aggregates AMF models to generate a regional model w'_d for global aggregation, and distributes the optimized model w_d to local AMFs after regional personalized optimization.

3) *AMF* S_a : AMF performs access and mobility management functions. In our system, AMF uses a local intelligent paging decision model, w_a , to send paging requests to target gNBs, thereby upgrading the rule-driven paging mechanism.

4) *gNB g*: gNB is the base station. In our system, the gNB sends paging requests to all UEs within its paging area according to instructions from the AMF.

5) *UE u*: UE refers to the terminal device used for accessing the network. In our system, UE handovers across gNBs are tracked by AMF. Through using RelTime-PRE, AMF predicts the future serving gNB for each UE, reducing paging overhead.

To align with the 3GPP vision of introducing intelligence into network management and service procedures [31], the proposed 6G-PFIP incorporates data-driven decision-making into the existing paging process as shown in Fig. 2. Specifically, the paging procedure in 6G-PFIP is described as follows:

- ① The UE initiates a paging request, indicating the requirement for service establishment.
- ② gNB-1 forwards the request to the distributed core network's AMF, where the UE context and related service information are retrieved.
- ③ Based on the UE context and historical mobility data, the distributed NWDAF performs intelligent inference to predict the most probable target cells or regions, enabling a data-driven paging decision instead of a rule-based broadcast.
- ④ The AMF delivers paging requests to the target core network according to the optimized decision results, thereby minimizing redundant signaling transmissions.
- ⑤ The target AMF configures the paging policy by incorporating the predicted location information, thereby narrowing the paging scope.
- ⑥ The AMF relays the paging request to gNB-2, which corresponds to the predicted target cell or region.
- ⑦ gNB-2 broadcasts the paging message within the selected coverage area, allowing the UE to respond and establish the connection.

Compared with the existing paging mechanism, which uses TA-level broadcasting and rule-based control, the proposed 6G-PFIP introduces intelligent, data-driven paging decisions through NWDAF-assisted prediction. This approach reduces signaling overhead and improves paging efficiency and adaptability. The procedure remains fully compliant with the 3GPP architecture [32]. It leverages standardized network functions and interfaces without altering the fundamental signaling framework.

B. Threat Model

This paper assumes that the NWDAF and D-NWDAFs in the 6G core network are honest and trustworthy. However, the communication links and participating clients may still be exposed to potential security threats.

(1)-Adversary Model:

The adversary's capabilities are defined as follows:

- The adversary can eavesdrop on the communication channels between AMFs and D-NWDAFs, intercepting transmitted information such as model parameters, gradients, and intermediate updates during the federated learning process.
- The adversary can perform model inference attacks (e.g., membership inference and model inversion) using intercepted parameters to infer sensitive UE information, such as mobility trajectories and location distributions.

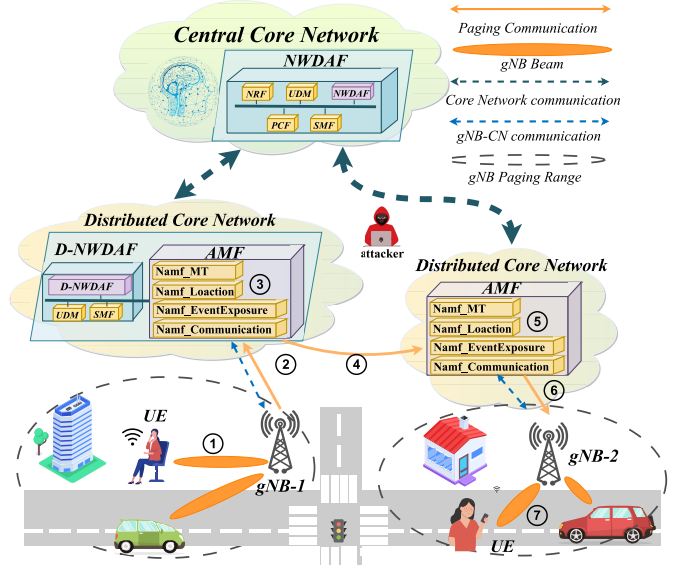


Fig. 2. The schematic illustrates the system model of 6G-PFIP.

- The adversary may act as a malicious client participating in the federated training process. Such an imposter can upload manipulated model updates or attempt to infer private information from the global model by repeatedly participating in the aggregation process.
- The adversary can perform replay or passive analysis attacks on the public communication channel, but cannot access raw UE data stored locally at each client.

(2)-Security Assumptions:

- The NWDAF and D-NWDAFs are trusted entities and follow the protocol correctly without leaking raw data.
- UE data remain locally stored on edge nodes (e.g., AMFs) and are not directly shared during training.

To defend against the above threats, we introduce local differential privacy (LDP) into the 6G-PFIP framework, enabling on-demand privacy protection, and analyze its privacy-preserving capability in Section VI.

C. Design Goals

Motivated by the challenges identified in the **Introduction** (Section I), this paper aims to develop an efficient, distributed-compatible, and privacy-preserving intelligent paging mechanism for future 6G networks. To address the aforementioned challenges, the proposed 6G-PFIP is designed to achieve the following three goals.

GOAL_1-High Efficiency. Rule-based paging mechanisms struggle to support large-scale access and highly dynamic mobility, resulting in excessive signaling overhead and paging latency. Therefore, accurate and adaptive paging decisions are essential for reducing the paging search space and improving resource utilization.

GOAL_2-Distributed-Compatible. Existing centralized paging mechanisms rely on network-wide information exchange between the RAN and the CN (Core Network), which limits scalability and introduces substantial communication

overhead. To support the distributed architecture of 6G networks, paging mechanisms should enable decentralized coordination and scalable deployment.

GOAL 3-Privacy Protection. Existing paging mechanisms generally lack dedicated privacy-preserving designs, despite involving sensitive UE information, such as user location and subscriber identity, during signaling. Moreover, the open and distributed architecture of 6G networks further increases the risk of privacy leakage. Therefore, the design of the paging mechanism should incorporate privacy-preserving measures.

IV. PAGING OPTIMIZATION

In the paging process, each UE connects to only one gNB at a time, so we use the connected gNB's coordinates for paging, assuming UEs always remain within a gNB's coverage area. Moreover, paging the TA/TAL after failing at the last-serving gNB is referred to as the second paging attempt. Our optimization objective is to reduce the total number of paging requests sent to gNBs during the second attempt. This objective aims to minimize paging signaling resource overhead Z , while maintaining high paging accuracy and complying with protocol constraints.

It can be expressed as:

$$Z = \sum_{i=1}^{n_a} \sum_{k=1}^{n_u} \sum_{g_j \in G_{a_i}} y_{a_i, u_k, g_j}, \quad (1)$$

where the binary variable $y_{a_i, u_k, g_j} \in Y^{n_a \times n_u \times n_g}$ indicates whether AMF a_i pages UE u_k via gNB g_j .

Notably, minimizing Z involves multiple constraints, and we explain them in detail below.

Constraint 1: Each UE is managed by one AMF during the paging process, represented by the first constraint below:

$$\sum_{i=1}^{n_a} z_{a_i, u_k} = 1, \quad \forall u_k \in U, \quad (2)$$

$$y_{a_i, u_k, g_j} \leq z_{a_i, u_k}, \quad \forall g_j \in G_{a_i}, \quad (3)$$

where the binary variable z_{a_i, u_k} represents whether UE u_k is assigned to AMF a_i , and the inequality ensures that only the responsible AMF can page its UE through managed gNBs.

Constraint 2: We assume each AMF has sufficient computing resources for paging. However, due to hardware constraints and to prevent service interference, gNB g_j has a maximum simultaneous paging capacity L_{g_j} . Thus, the second constraint is shown as:

$$\sum_{i=1}^{n_a} \sum_{k=1}^{n_u} y_{a_i, u_k, g_j} \leq L_{g_j}, \quad \forall g_j \in G_{a_i}. \quad (4)$$

Constraint 3: We prioritize paging gNBs near the UE's predicted location to reduce signaling overhead. So the third constraint limiting concurrent paging to K gNBs, expressed as:

$$\sum_{g_j \in G_{a_i}} y_{a_i, u_k, g_j} \leq K, \quad \forall u_k \in U, \exists a_i \in A. \quad (5)$$

Constraint 4: The fourth constraint ensures at least one paging attempt per UE during the paging process, and is formulated as:

$$\sum_{g_j \in G_{a_i}} y_{a_i, u_k, g_j} \geq 1, \quad \forall u_k \in U, \exists a_i \in A. \quad (6)$$

Constraints 3 and 4 require that each paging request from a_i to u_k must target between 1 and K gNBs. The process escalates to TA if unsuccessful after paging K gNBs.

We propose RelTime-PRE for paging decisions, deployed on AMFs to get the paging success probability p_{a_i, u_k} . RelTime-PRE uses UE handover history to predict locations and pages the nearest gNB. The paging success probability p_{a_i, u_k} depends on whether the actual serving gNB of UE u_k is among the K nearest gNBs, expressed as:

$$p_{a_i, u_k} = \frac{\sum_{u \in U_{a_i}} I[g_u^{label} \in \text{Rel-P}_K((x_u, y_u)_{pred})]}{|U_{a_i}|}, \quad (7)$$

where U_{a_i} is the UE set of AMF a_i , $(x_u, y_u)_{pred}$ is the predicted coordinates of UE u , g_u^{label} is the actual serving gNB of UE u . The function $\text{Rel-P}_K((x_u, y_u)_{pred})$ returns the K nearest gNBs, with $I[\cdot] = 1$ if the actual serving gNB is in this set and 0 otherwise.

To facilitate the selection of candidate paging gNB, the predicted paging probability for u_k is allocated across p_{a_i, u_k, g_j} for $n_{g_{a_i}}$ gNBs. Thus, when AMF a_i pages UE u_k through gNB g_j , the paging failure probability is expressed as:

$$p_{a_i, u_k}^{\text{fail}} = 1 - \sum_{j=1}^{n_{g_{a_i}}} p_{a_i, u_k, g_j} y_{a_i, u_k, g_j}, \quad (8)$$

where y_{a_i, u_k, g_j} indicates whether g_j is selected to page u_k under a_i .

Constraint 5: To ensure paging reliability, we constrain the average paging failure probability across all AMFs and UEs by a threshold P^{fail} , leading to the fifth constraint as follows:

$$\frac{1}{n_u} \sum_{i=1}^{n_a} \sum_{k=1}^{n_u} p_{a_i, u_k}^{\text{fail}} \leq P^{\text{fail}}, \quad (9)$$

which aims to balance paging efficiency while avoiding excessive latency from aggressive signaling overhead reduction.

Combining the above constraints, the overall optimization problem can be formulated as:

$$\begin{aligned} \arg \min_Y \quad & Z = \sum_{i=1}^{n_a} \sum_{k=1}^{n_u} \sum_{g_j \in G_{a_i}} y_{a_i, u_k, g_j}, \\ \text{s.t.} \quad & \begin{cases} \sum_{i=1}^{n_a} z_{a_i, u_k} = 1, \forall u_k \in U \\ y_{a_i, u_k, g_j} \leq z_{a_i, u_k}, \forall a_i \in A, \forall g_j \in G_{a_i} \\ \sum_{i=1}^{n_a} \sum_{k=1}^{n_u} y_{a_i, u_k, g_j} \leq L_{g_j}, \forall g_j \in G_{a_i} \\ \sum_{g_j \in G_{a_i}} y_{a_i, u_k, g_j} \geq 1, \forall u_k \in U, \exists a_i \in A \\ \sum_{g_j \in G_{a_i}} y_{a_i, u_k, g_j} \leq K, \forall u_k \in U, \exists a_i \in A \\ \frac{1}{n_u} \sum_{i=1}^{n_a} \sum_{k=1}^{n_u} p_{a_i, u_k}^{\text{fail}} \leq P^{\text{fail}}. \end{cases} \end{aligned} \quad (10)$$

To analyze the computational complexity of Problem (10), we apply three simplifications. First, we restrict the number of AMFs to $n_a = 1$, which removes the index i and eliminates Constraint 1, thereby simplifying the decision variable to

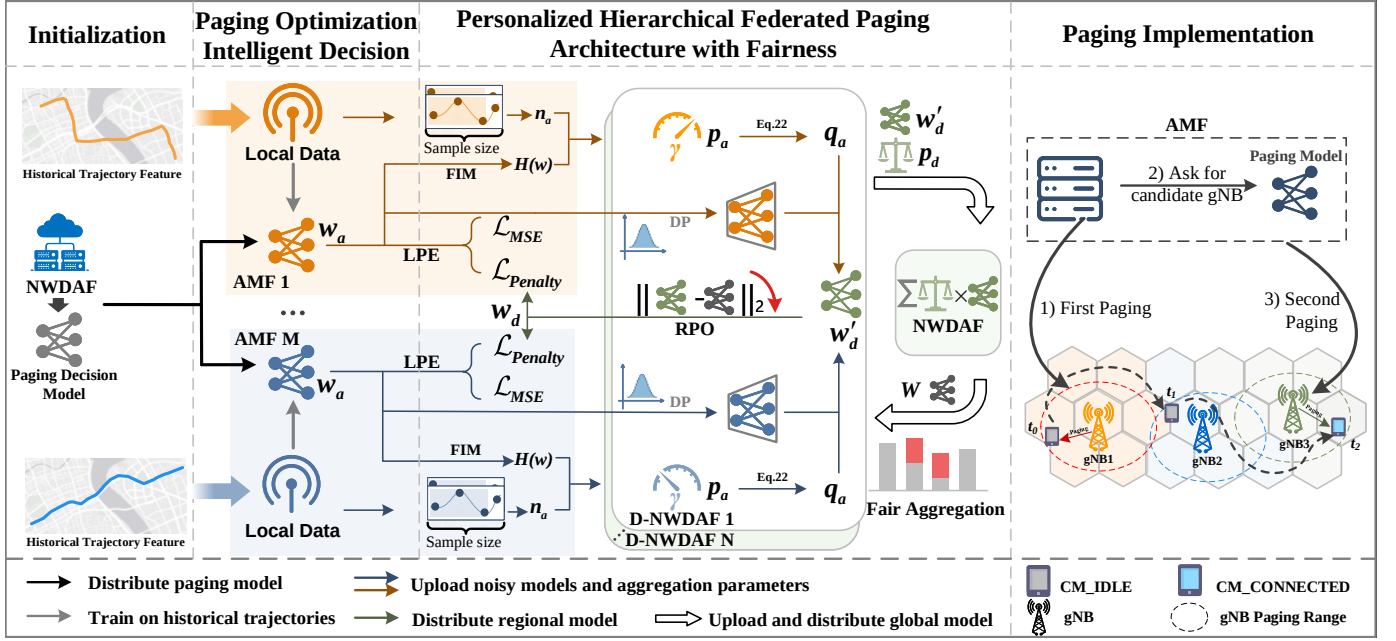


Fig. 3. The schematic diagram of 6G-PFIP.

y_{u_k, g_j} . Second, we set the concurrent paging limit K to 1, effectively merging Constraints 3 and 4 into the single equation $\sum_{g_j} y_{u_k, g_j} = 1$. Finally, the average paging failure threshold P^{fail} is set to 1, rendering Constraint 5 inactive.

Consequently, the original paging optimization problem reduces to the following form:

$$\begin{aligned}
 & \arg \min_Y \sum_{u_k} \sum_{g_j} y_{u_k, g_j}, \\
 & \text{s.t.} \sum_{g_j} y_{u_k, g_j} = 1, \quad \forall u_k, \\
 & \sum_{u_k} y_{u_k, g_j} \leq L_{g_j}, \quad \forall g_j, \\
 & y_{u_k, g_j} \in \{0, 1\}.
 \end{aligned} \tag{11}$$

To optimize paging in this context, we propose an intelligent paging mechanism named 6G-PFIP. This mechanism is equipped with Paging Optimization Intelligent Decision to improve the quality of paging services and adapt to the architecture of next-generation mobile communication networks. Furthermore, privacy-preserving capabilities can be flexibly provided according to system requirements.

V. OUR PROPOSED 6G-PFIP

As shown in Fig. 3, the 6G-PFIP execution process comprises four strategies, including Initialization, Paging Optimization Intelligent Decision, HFFA, and Paging Implementation. The main notations and their meanings for 6G-PFIP are summarized in Table I.

A. Initialization

During initialization, the paging process involves AMFs, gNBs, and UEs. HFFA involves three entities: NWDAF, D-NWDAF, and AMF. NWDAF initializes and distributes a paging intelligent decision model to D-NWDAFs, which forward

it to AMFs. It results in a three-level structure: a global model W at NWDAF, a regional model w_d at D-NWDAFs, and a local model w_a at AMFs.

B. Paging Optimization Intelligent Decision

To minimize paging signaling overhead, we design an intelligent decision model to optimize paging. Its role in 6G-PFIP is illustrated in Fig. 3. The paging optimization intelligent decision module contains the core processing model denoted as RelTime-PRE. The architecture of RelTime-PRE is shown in Fig. 4. The detailed RelTime-PRE process consists of four steps, as follows.

Step 1—UE Data Formation Input: The UE trajectory data of length L_d can be represented as $x_i^h = \{x_i^{h,1}, x_i^{h,2}, \dots, x_i^{h,L_d}\}$, where $h \in \{1, 2\}$ denotes two trajectory channels corresponding to gNB longitude and latitude, respectively. Each channel is trained independently to avoid feature interference, and the extracted representations are subsequently fused. For each UE, the AMF provides its gNB handover sequence $x_i \in \mathbb{R}^{2 \times L_d}$ as the model input. The output of RelTimeNet is the predicted UE trajectory $\hat{y}_i \in \mathbb{R}^{2 \times L_p}$, where L_p denotes the prediction length.

Step 2—Trajectory Segmentation: Extracting semantic information from a single sequence is challenging. Therefore, for the UE trajectory data x_i^h , we divide it into N_s trajectory segments of length L_s with a sliding stride S , as follows:

$$N_s = \left\lfloor \frac{L_d - L_s}{S} \right\rfloor + 1. \tag{12}$$

The m -th trajectory segment is defined as:

$$z_m^h = \{x_i^{h, (m-1)S+1}, \dots, x_i^{h, (m-1)S+L_s}\}, \tag{13}$$

where $m \in \{1, 2, \dots, N_s\}$. Trajectory segmentation captures local temporal trends and periodic mobility patterns.

TABLE I
KEY NOTATIONS AND MEANINGS FOR 6G-PFIP

Notation	Description
Key Components and Mechanisms:	
6G-PFIP	A federated intelligent paging mechanism for 6G networks.
RelTime-PRE	An intelligent paging decision model in 6G-PFIP.
HFWA	A personalized federated paging architecture in 6G-PFIP.
RPO	A regional personalized optimization mechanism in HFWA.
LPE	A local personalization enhancement mechanism in HFWA.
FWA	A fair weighted aggregation mechanism in HFWA.
PPE	A Privacy Protection Enhancement in HFWA.
Paging Phase:	
a_i, A	Set of AMFs
g_j, G_{a_i}	Set of gNBs within the paging area of AMF a_i
u_k, U	Set of UEs
U_{a_i}	Set of UEs managed by AMF a_i
Z	Paging signaling overhead
n_a	Number of AMFs
$n_{g_{a_i}}$	Number of gNBs in the paging area of AMF a_i
n_u	Number of UEs
L_{g_j}	Max simultaneous paging requests supported by a gNB
K	Max number of gNBs paged at once
(x_u, y_u)	Predicted coordinates of UE u from the paging model
g_u^{label}	Ground-truth gNB of the UE u
$Rel - P_K()$	KD-Tree mapping of predicted location to K nearest gNBs
p_{a_i, u_k}	Success probability of AMF a_i paging UE u_k
p_{a_i, u_k, g_j}	Success probability of AMF a_i paging UE u_k via gNB g_j
p_{a_i, u_k}^{fail}	Failure probability of AMF a_i paging UE u_k
P_{fail}^{th}	Threshold for average paging failure rate
HFWA Phase:	
x_i^h	Handover history of UE across different gNBs
y_i^h	Predicted UE location from the paging decision model
$\mathcal{L}_{MSE}$	Mean squared error between predictions and ground truth
$\mathcal{L}_{penalty}$	The squared L_2 norm between local and regional models
h_a	The importance of model parameters
p_a	The multi-scale weight of the local model
q_a	The multi-scale aggregation weight for AMF
q_d	The multi-scale aggregation weight for D-NWDAF
n_a	Number of samples held by AMF
S_a	Set of all AMFs
S_d	Set of all D-NWDAFs
S_{d_a}	Set of AMFs managed by D-NWDAF d_a
T	The total number of communication rounds
l_1	Local training epochs
l_2	Edge aggregation epochs
w_a^l	Model parameters of AMF a after l rounds of local training
$\tilde{w}_a^t$	Noised local model from AMF at round t
w_d^t	The regional model aggregated by D-NWDAF in round t
W^t	The global model aggregated by NWDAF in round t
w_d^t	Optimized regional model at D-NWDAF in round t

Step 3—RelTimeNet: RelTimeNet adopts a two-stage temporal modeling strategy. First, each trajectory segment is projected into a latent embedding space:

$$e_m = \phi(z_m), \quad (14)$$

where $\phi(\cdot)$ denotes the segment embedding function.

The embedded segment sequence $E = \{e_1, e_2, \dots, e_{N_s}\}$ is then processed by the Transformer Encoder to capture global temporal dependencies across trajectory segments:

$$H_T = \text{Transformer}(E). \quad (15)$$

Subsequently, BiLSTM further models bidirectional temporal evolution over the Transformer representations:

$$H_B = \text{BiLSTM}(H_T). \quad (16)$$

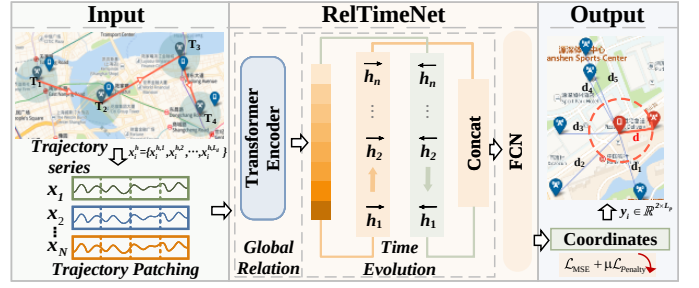


Fig. 4. The schematic diagram of the RelTime-PRE model.

Compared with relying solely on the attention mechanism, the introduced BiLSTM provides additional temporal regularization and improves the modeling of sequence continuity. The Mean Squared Error (MSE) is adopted to quantify the prediction error between the ground-truth trajectory y_i and the predicted trajectory $\hat{y}_i$, as follows:

$$\mathcal{L} = \frac{1}{2n_u} \sum_{i=1}^{n_u} \sum_{h=1}^2 \|\hat{y}_i^h - y_i^h\|_2^2. \quad (17)$$

Step 4—Paging Candidate Selection & Accuracy Calculation: The KD-tree is employed to organize gNB locations in a two-dimensional spatial domain defined by longitude and latitude. This structure formulates the paging candidate selection problem as a K -Nearest Neighbor (KNN) search. Given the predicted UE coordinate $\hat{y}_i$, the KD-tree efficiently retrieves the K nearest gNBs using Euclidean distance. During the search process, subtrees whose minimum distance exceeds the current search radius are pruned to reduce computational complexity.

If the target gNB is included in the K nearest gNBs associated with the predicted UE coordinate $\hat{y}_i$, the paging attempt is regarded as successful. The overall paging accuracy is defined as:

$$Acc = \frac{1}{n_u} \sum_{i=1}^{n_u} I(y_i \in \text{KD-T}_K(\hat{y}_i)), \quad (18)$$

where $I(\cdot) = 1$ indicates a successful paging attempt, and 0 otherwise.

C. HFWA

To enable efficient, privacy-preserving use of RelTime-PRE across distributed AMFs, we design a Personalized Hierarchical Federated Paging Architecture with Fairness (HFWA). It's four blocks, as shown in Fig. 5, effectively addresses the challenges posed by Non-IID UE data, regional heterogeneity, unfair aggregation, and the increasing demand for privacy preservation in 6G networks.

Block 1—Regional Personalized Optimization (RPO): To address regional UE data heterogeneity, HFWA incorporates the regional personalized optimization mechanism, which operates during the D-NWDAF process. The central NWDAF aggregates all regional models to generate a global model W^t , and distributes it to all D-NWDAFs. Each D-NWDAF optimizes its regional model w_d^t based on W^t to blend regional and global knowledge into the optimized model w_d^{t+1} ,

subsequently distributed to AMFs. The detailed procedure is as follows:

$$w_d^{t+1} \leftarrow w_d^{tt} - \lambda(w_d^{tt} - W^{t+1}), \quad (19)$$

where λ controls the approximation rate every communication round t . We set different λ for D-NWDAFs, allowing the regional models to maintain personalization while incorporating global knowledge to reduce model drift.

Block 2—Local Personalization Enhancement (LPE): To handle AMF-level UE mobility heterogeneity, we introduce a penalty term to the AMF loss function and operate during the AMF process. The penalty term aims to minimize the distance between the optimized model w_d^t and the local model w_a^l , enabling each AMF to balance regional and local knowledge for better prediction performance. The loss function is expressed as follows:

$$\mathcal{L}_{w_a^{l+1}} = \mathcal{L}_{MSE} + \mu \mathcal{L}_{penalty}, \quad (20)$$

$$\mathcal{L}_{MSE} = \frac{1}{2} \sum_{h=1}^2 \|\hat{y}_i^h - y_i^h\|_2^2, \quad (21)$$

$$\mathcal{L}_{Penalty} = \frac{1}{2} \|w_a^l - w_d^t\|_2^2, \quad (22)$$

where μ is a balancing factor that controls the distance between the local and regional models.

Block 3—Fair Weighted Aggregation (FWA): To fully leverage the data knowledge of each AMF, we build a fair weighted aggregation method mechanism with the multi-scale aggregation weights for AMF q_a and for D-NWDAF q_d . It operates during the AMF, D-NWDAF, and NWDAF processes.

Fisher Information Matrix (FIM) quantifies the sensitivity of the model with respect to parameter variations by computing the squared gradient of the log-likelihood function [33], [34]. From a geometric perspective, the FIM captures the local curvature of the loss landscape. Larger values indicate that small parameter perturbations lead to larger changes in the loss function. Therefore, parameters with higher Fisher information are considered more important because they exert a stronger influence on the model output.

Specifically, the FIM for parameter w_a is defined as:

$$H(w_a) = (\nabla_{w_a} \mathcal{L}_a(x_i, y_i, w_a))(\nabla_{w_a} \mathcal{L}_a(x_i, y_i, w_a))^T, \quad (23)$$

where $\nabla_{w_a} \mathcal{L}_a(x_i, y_i, w_a)$ denotes the gradient of the loss function with respect to parameter w_a .

To obtain an overall importance metric for the local model, we aggregate the Fisher information across all parameters as follows:

$$h_a = \sum_{v=1}^V H(w_a^v), \quad (24)$$

where V is the total number of parameters in the local model, and $H(w_a^v)$ represents the Fisher information associated with the v -th parameter.

The aggregated importance metric h_a reflects the cumulative sensitivity of the local model parameters in client a and is subsequently used in Eqs. (19)-(20) to evaluate and compare parameter importance during the optimization process. Parameters with larger Fisher information values are considered more

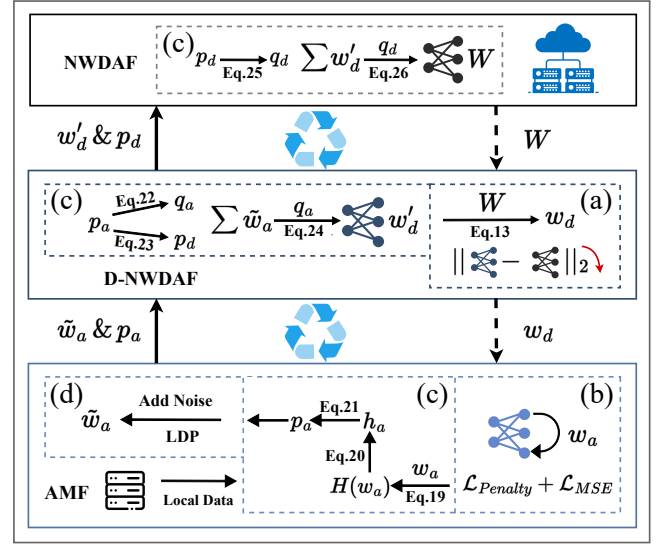


Fig. 5. The schematic diagram of the HFPA. It includes four blocks: (a) Regional Personalized Optimization operates during the D-NWDAF process. (b) Local Personalization Enhancement operates during the AMF process. (c) Fair Weighted Aggregation operates during the AMF, D-NWDAF, and NWDAF processes. (d) Privacy Protection Enhancement Mechanism operates during the AMF process.

influential and thus contribute more significantly to the model behavior.

D-NWDAF uses the received AMF models w_a and weight p_a to aggregate the regional model w_d^t as follows:

$$q_a = p_a / \sum_{i \in S_{d_a}} p_i, \quad (25)$$

$$p_d = \sum_{i \in S_{d_a}} p_i, \quad (26)$$

$$w_d^{tt} = \sum_{a \in S_{d_a}} q_a \times w_a^t, \quad (27)$$

where q_a is the multi-scale aggregation weight for AMF, p_d represents the multi-scale weight for D-NWDAF.

Subsequently, based on regional model w_d^t and weight p_a , NWDAF aggregates the global model W as follows:

$$q_d = p_d / \sum_{i \in S_d} p_i, \quad (28)$$

$$W^{t+1} = \sum_{d \in S_d} q_d \times w_d^{tt}, \quad (29)$$

where S_d represents D-NWDAF set within a NWDAF.

Block 4—Privacy Protection Enhancement (PPE): To mitigate privacy leakage in open 6G network environments, we introduce local differential privacy (LDP) [35] into the 6G-PFIP framework during the AMF process. Specifically, Gaussian noise is added to local model parameters before regional uploads to mask sensitive information contained in UE data.

The proposed framework does not adopt a fixed-level or uniform privacy-protection strategy. Instead, it provides on-demand privacy protection capability. One reason existing

paging mechanisms lack privacy-preserving capabilities is the concern that this would sacrifice paging performance. However, this is clearly insufficient in scenarios with high privacy requirements. Therefore, 6G-PFIP aims to balance paging performance and privacy preservation, enabling on-demand privacy protection by adapting the privacy level to varying paging requirements.

From the perspective of differential privacy, the strength of privacy protection is governed by the privacy parameters (ϵ, δ) . These parameters are directly related to the noise scale σ in the Gaussian mechanism. Therefore, on-demand privacy protection can be achieved by dynamically adjusting the noise scale σ according to service requirements:

$$\sigma \propto \frac{\Delta f}{\epsilon}. \quad (30)$$

A smaller ϵ corresponds to stronger privacy protection. This requires a larger noise scale σ . In contrast, a larger ϵ allows for reduced noise injection and improved paging performance. By adaptively adjusting σ (and equivalently (ϵ, δ)), the proposed framework enables flexible trade-offs between privacy preservation and paging efficiency across different application scenarios. To formally characterize the above privacy-preserving mechanism, we introduce the following definitions of differential privacy and its key components.

Definition 1 (Adjacent Dataset): D and D' are two adjacent datasets with the same structure, and $|D \Delta D'| = 1$. Formally, $D \Delta D'$ denotes the symmetric difference, and $|D \Delta D'|$ represents the number of different elements between them.

Definition 2 (Generalized Differential Privacy): A randomized mechanism $\mathcal{A} : \mathcal{D} \rightarrow \mathcal{R}$ satisfies (ϵ, δ) -DP, where privacy parameters $\epsilon > 0$ and $0 \leq \delta \leq 1$. For any two local adjacent datasets $D, D' \in \mathcal{D}$ and any subset of outputs $\mathcal{O} \subseteq \mathcal{R}$, the following condition is satisfied:

$$Pr[\mathcal{A}(D) \in \mathcal{O}] \leq e^\epsilon Pr[\mathcal{A}(D') \in \mathcal{O}] + \delta, \quad (31)$$

where $\mathcal{D}$ is the domain of data and $\mathcal{R}$ is the range of outputs. ϵ is the privacy budget; the smaller its value, the stronger the privacy protection provided. δ represents the upper bound on the probability of violating the strict (ϵ, δ) -DP constraint.

Definition 3 (L_2 -sensitivity): f is the query function, and the L_2 -sensitivity Δf can be defined as:

$$\Delta f = \max_{D, D'} \|f(D') - f(D)\|_2. \quad (32)$$

Definition 4 (Gaussian Mechanism): Gaussian noise $Y \sim N(0, \sigma^2)$ is added to achieve (ϵ, δ) -DP, as follows:

$$\mathcal{A}(D') = f(D') + N(0, \sigma^2), \quad (33)$$

where the noise scale σ is $\sigma = c\Delta f/\epsilon$, $c = \sqrt{2\ln(1.25/\delta)}$.

In HFFPA, we clip the parameters of each AMF model to ensure they lie within the range $[-C, C]$ as follows:

$$\bar{w}_a^{l_1} \leftarrow \frac{w_a^{l_1}}{\max(1, \|w_a^{l_1}\|_2/C)}. \quad (34)$$

Each AMF perturbs its local model parameters by adding Gaussian Noise as follows:

$$\tilde{w}_a^t \leftarrow \bar{w}_a^{l_1} + N(0, (\sigma^{l_1})^2), \quad (35)$$

where $\sigma^{l_1} \geq 2Cl_1\sqrt{2\ln(1.25/\delta)/(\epsilon|D'|)}$, we proof it in Theorem 2. We set $\sigma^{l_1} = 2Cl_1\sqrt{2\ln(1.25/\delta)/(\epsilon|D_{min}|)}$ and D_{min} is the AMF with the smallest number of gNBs.

The HFFPA consists of three steps, including the AMF process, the D-NWDAF process, and the NWDAF process. Its process is shown in Algorithm 1.

Algorithm 1 Personalized Hierarchical Federated Paging Architecture with Fairness

Input: $T, l_1, l_2, n_a, n_d, n, \eta, \mu, \gamma, S_{d_a}$, and S_d .

Output: Local personalized model $w_a^{l_1}$ for each AMF.

```

1: for round  $t \in \{1, 2, \dots, T\}$  do
2:   AMF process:
3:   Receive  $w_d^t$  from its corresponding D-NWDAF
4:   for local updates  $l \in \{0, 1, \dots, l_1 - 1\}$  do
5:     Perform local personalization update:
6:      $w_a^{l+1} \leftarrow w_a^l - \eta(\nabla \mathcal{L}_{MSE} + \mu \nabla \mathcal{L}_{penalty})$ 
7:   end for
8:   Calculate the parameters importance  $h_a$ :
9:    $h_a \leftarrow \sum_{v=1}^V H(w_a^v)$ 
10:  Calculate the AMF multi-scale weight  $p_a$ :
11:   $p_a \leftarrow (1 - \gamma)n_a / \sum_{i \in S_{d_a}} n_i + \gamma h_a / \sum_{i \in S_{d_a}} h_i$ 
12:  Gradient clipping:
13:   $\bar{w}_a^{l_1} \leftarrow w_a^{l_1} / \max(1, \|w_a^{l_1}\|_2/C)$ 
14:  Add noise:
15:   $\tilde{w}_a^t \leftarrow \bar{w}_a^{l_1} + N(0, (\sigma_n^{l_1})^2)$ 
16:  Upload  $\tilde{w}_a^t$  and  $p_a$  to its corresponding D-NWDAF
17:  D-NWDAF process:
18:  Receive  $\tilde{w}_a^t$  and  $p_a$  from its managed AMFs
19:  D-NWDAF aggregate AMF model parameters:
20:   $q_a \leftarrow p_a / \sum_{i \in S_{d_a}} p_i$ 
21:   $w_d^t \leftarrow \sum_{a \in S_{d_a}} q_a \times w_a^t$ 
22:  Calculate the NWDAF multi-scale weight  $p_d$ :
23:   $p_d \leftarrow \sum_{i \in S_{d_a}} p_i$ 
24:  if  $t \% l_2 == 0$  then
25:    Send  $w_d^t$  and  $\sum_{i \in S_{d_a}} q_i$  to NWDAF
26:    Receive global model  $W^{t+1}$  from NWDAF
27:    Perform regional personalization update:
28:     $w_d^{t+1} \leftarrow w_d^t - \lambda(w_d^t - W^{t+1})$ 
29:  else
30:    Compute the regional model:
31:     $w_d^{t+1} \leftarrow w_d^t$ 
32:  end if
33:  Distribute  $w_d^{t+1}$  to all AMFs under its domain
34:  NWDAF process:
35:  Receive  $w_d^t$  and  $p_d$  from all D-NWDAFs
36:  Aggregate to get the global model  $W^{t+1}$ :
37:   $q_d \leftarrow p_d / \sum_{i \in S_d} p_i$ 
38:   $W^{t+1} \leftarrow \sum_{d \in S_d} q_d \times w_d^t$ 
39:  Broadcast  $W^{t+1}$  to all D-NWDAFs
40: end for
```

D. Paging Implementation

In our paging mechanism, paging implementation comprises four steps, including first paging, intelligent paging, escalation to TA/TAL, and completion & reset.

Step 1—First Paging: When AMF detects new data arrival for a UE in CM-IDLE mode, the system immediately initiates paging to the last-serving gNB according to 3GPP. If no response is received within T_{succ} , the process advances to the next step.

Step 2—Intelligent Paging: During intelligent paging, AMF uses UE handover data $x_i^h = \{x_i^{h,1}, x_i^{h,2}, \dots, x_i^{h,L_d}\}$ to predict the UE location coordinates $(x_u, y_u)_{pred}$ through RelTime-PRE. AMF then queries KD-Tree for K nearest gNBs and pages them, succeeding if any gNB gets a response within T_{succ} .

Step 3—Escalation to TA/TAL: If all K candidate gNBs fail, AMF expands paging to the entire TA and subsequently to the complete TAL. During this step, paging requests are broadcast to all gNBs until the UE is successfully located.

Step 4—Completion & Reset: After successful paging, AMF records the responding gNB for signaling and terminates the process by resetting timers until the next paging event.

6G-PFIP pioneers the integration of rule-based paging with a secondary paging attempt. It reduces TA/TAL broadcasts while maintaining high success rates and lowering both paging latency and signaling overhead.

VI. PRIVACY ANALYSIS

In Section V, we apply LDP to local model parameters when uploading them to D-NWDAF. To evaluate the effectiveness of LDP in privacy preservation, we provide the privacy-preserving capability analysis for 6G-PFIP in this section.

To protect the privacy of parameters, each AMF perturbs its model parameters with Gaussian noise, satisfying (ϵ, δ) -DP. In the following, we introduce the theorems. Theorem 1 analyzes the bound of sensitivity Δf . Theorem 2 provides the range of noise scale σ^{l_1} under a 6G distributed core network in this paper.

Theorem 1: For $\forall d_u \in D'$, it holds $\|w^{d_u}\|_2 \leq C$, the sensitivity Δf is constrained by $\Delta f \leq \frac{2C}{|D'|}$.

Proof: For $\forall d_u \in D'$, there are two adjacent UE trajectory datasets. We denote them as D and $D' = D \cup \{d_u\}$, where $|D'| = |D| + 1$.

The model parameters $w^{D'}$ of D' can be decomposed as:

$$w^{D'} = \frac{|D|w^D + w^{d_u}}{|D'|}, \quad (36)$$

where w^D is the trained model parameters of the UE trajectory dataset D .

We define $f(D) = w^D$, so the sensitivity Δf in 6G-PFIP can be written as $\Delta f = \max_{D, D'} \|w^{D'} - w^D\|_2$.

The difference between the parameters of two AMF models $w^{D'} - w^D$, can be expressed as:

$$w^{D'} - w^D = \frac{w^{d_u} - w^D}{|D'|}. \quad (37)$$

Therefore, $\|w^{D'} - w^D\|_2$ is calculated as follows:

$$\begin{aligned} \|w^{D'} - w^D\|_2 &= \frac{1}{|D'|} \|w^{d_u} - w^D\|_2 \\ &\leq \frac{1}{|D'|} (\|w^{d_u}\|_2 + \|w^D\|_2). \end{aligned} \quad (38)$$

For any $d_u \in D'$ holds $\|w^{d_u}\|_2 \leq C$, so the range of $\|w^D\|_2$ can be expressed as:

$$\begin{aligned} \|w^D\|_2 &= \left\| \frac{1}{|D|} \sum_{d_i \in D} w^{d_i} \right\|_2 \\ &\leq \frac{1}{|D|} \sum_{d_i \in D} \|w^{d_i}\|_2 \leq C. \end{aligned} \quad (39)$$

Thus, $\|w^{d_u}\|_2 + \|w^D\|_2 \leq C + C = 2C$.
 $\|w^{D'} - w^D\|_2$ can be updated as:

$$\|w^{D'} - w^D\|_2 \leq \frac{2C}{|D'|}. \quad (40)$$

The sensitivity range Δf in 6G-PFIP is given by:

$$\Delta f = \max_{D, D'} \|f(D') - f(D)\|_2 \leq \frac{2C}{|D'|}. \quad (41)$$

Theorem 1 is proved.

Remark 1: We assume w^D and $w^{D'}$ comply with the aggregation characteristics of HFPFA.

Theorem 2: To ensure (ϵ, δ) -DP for each AMF in the uplink, and local epochs of AMF are l_1 , $\sigma^{l_1} \geq \frac{2Cl_1\sqrt{2\ln(1.25/\delta)}}{\epsilon|D'|}$.

Proof: According to Theorem 1, the sensitivity of a single AMF update satisfies: $\Delta f \leq \frac{2C}{|D'|}$.

After l_1 AMF local training iterations, the cumulative sensitivity grows linearly and can be expressed as:

$$\|w_{l_1}^{D'} - w_{l_1}^D\|_2 \leq l_1 \cdot \frac{2C}{|D'|}. \quad (42)$$

Based on Definition 4, the local noise scale σ^{l_1} is given by:

$$\sigma^{l_1} \geq \frac{2Cl_1\sqrt{2\ln(1.25/\delta)}}{\epsilon|D'|}. \quad (43)$$

Theorem 2 is proved.

VII. EVALUATION

In this section, we evaluate the performance of 6G-PFIP from three perspectives: intelligent decision models for paging optimization, HFPFA, and its overall performance.

A. Experimental Setting

Dataset: We utilized the UE mobility trajectory dataset provided by Shanghai Telecom [36] to validate our mechanism. The dataset comprises 7.2 million access events generated by 9,481 mobile devices across 3,233 gNBs. Each record contains a timestamp (month, day, start time, and end time), the latitude and longitude of the corresponding gNB, and a unique user equipment identifier. Notably, the dataset naturally includes users with diverse activity frequencies and mobility patterns, which introduces inherent heterogeneity at the data level. To further increase heterogeneity, we apply Dirichlet-based partitioning. As shown in Fig. 6, each distributed node processes user data with distinct characteristics, reflecting the spatial and behavioral heterogeneity of UEs across different paging regions. To build on this and enable controllable, reproducible heterogeneity levels in our experiments, we further apply a

Dirichlet-based data partitioning strategy to distribute UE data across AMFs.

Baselines: To evaluate the performance of the paging intelligent decision model RelTime-PRE, we compare it with five widely used state-of-the-art intelligent models:

- **Long Short-Term Memory (LSTM)** [37]: LSTM employs gates to mitigate vanishing gradients, enabling modeling of long-term sequence dependencies.
- **FCN** [38]: A standard architecture for time-series feature extraction, sequentially stacks convolution, normalization, and ReLU layers.
- **LSTM-FCN** [39]: LSTM-FCN integrates LSTM and FCN to capture long-term and local patterns, demonstrating strong performance in time-series prediction.
- **ResCNN** [40]: ResCNN embeds residual modules in the first three convolutional layers, introducing residual-network mechanisms into CNN.
- **TST** [41]: A Transformer-based model uses multi-head attention to capture long-range dependencies and temporal interactions.

To assess the superiority of HFPA in real paging scenarios, we compare our scheme with four FL approaches that are widely adopted in the 6G FL:

- **FedAvg** [42]: FedAvg performs local training on clients and aggregates on the server to form a global model.
- **FedPer** [43]: FedPer shares base-layer parameters for aggregation while keeping personalized layers locally to handle client-specific heterogeneity.
- **FedProx** [44]: FedProx adds a proximal term to limit the distance between local and global models, addressing convergence issues from data heterogeneity.
- **Ditto** [45]: Ditto applies FedAvg for global aggregation, while clients use a regularizer to refine personalized models, balancing global alignment and local specificity.

Performance Metrics: We adopt two primary metrics:

- **Mean Squared Error (MSE):** We quantify the convergence of RelTime-PRE and HFPA by computing the MSE between the actual serving gNB and the predicted gNB.
- **Accuracy:** We evaluate the performance of 6G-PFIP by calculating paging accuracy and comparing paging latency and signaling overhead with existing methods.

Implementation: We deploy a total of 100 AMFs and group every 10 AMFs into 1 D-NWDAF for edge aggregation. The Dirichlet-based partitioning is employed to simulate heterogeneous user distributions across AMFs, reflecting the spatial and traffic diversity in practical paging scenarios. Specifically, for each user class k , the data distribution over M AMFs is sampled from a Dirichlet distribution:

$$(p_1^k, p_2^k, \dots, p_M^k) \sim \text{Dir}(\alpha), \quad (44)$$

where p_m^k denotes the proportion of class- k data assigned to the m -th AMF, and α is the concentration parameter controlling the degree of statistical heterogeneity.

The level of heterogeneity among AMFs is governed by α , where a larger α indicates a more uniform distribution, and a smaller α leads to highly skewed data allocation, with each AMF dominated by a limited subset of user patterns [46]–[48].

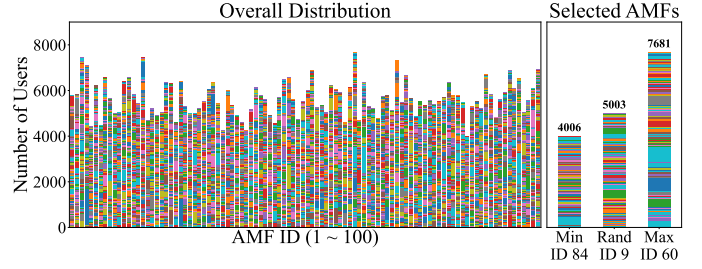


Fig. 6. The Heterogeneity of the Shanghai Telecom Dataset.

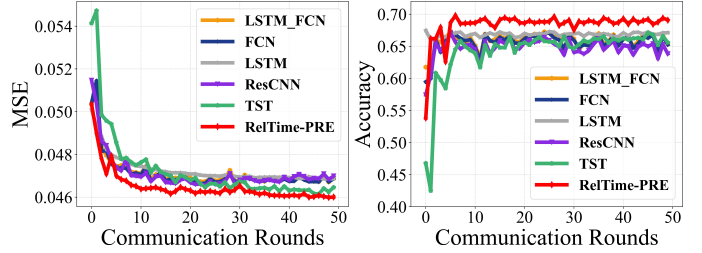


Fig. 7. Performance Comparison Between RelTime-PRE and Other Models.

The number of users assigned to each AMF ranges from 4,006 to 7,681, as illustrated in Fig. 6, where different colors denote distinct classes.

The total number of communication rounds is $T = 100$. The learning rate is $\eta = 0.001$, the balancing factor is $\mu = 0.01$, and the approximation rate is $\gamma = 0.02$. It is worth noting that our study is conducted under the ideal bandwidth assumption, in which all AMFs are assumed to transmit synchronously. Additionally, we set the KD-Tree parameter $K = 5$.

Experimental Environment: All experiments are implemented on a hardware platform consisting of three physical hosts and two Linux servers, equipped with NVIDIA A800 and RTX 5060TI GPUs, respectively. Experimental results are simulated and analyzed using PyTorch and MATLAB.

B. Experimental Results

1) *Effectiveness of RelTime-PRE under 6G UE Prediction Scenarios:* We evaluate the performance of RelTime-PRE against several state-of-the-art intelligent models in two metrics on the test set, as shown in Fig. 7.

As the number of communication rounds increases, RelTime-PRE consistently outperforms other models across all metrics. On the test set, all models show varying degrees of fluctuation. However, RelTime-PRE shows reduced fluctuation. Specifically, at the 36th round, it achieves 69.40% accuracy and 0.0461 MSE, with accuracy improvements of 4.07%, 4.32%, 2.44%, 4.99%, and 2.86% over the other models. The effectiveness of RelTime-PRE stems from two key factors. Trajectory Segmentation divides continuous trajectories into segments to better capture local movement patterns. Meanwhile, RelTimeNet fully exploits temporal dependencies via multi-head attention and BiLSTM, which model spatial correlations and temporal continuity, respectively, producing a more discriminative mobility representation.

2) *Evaluation of HFPA under IID Data Distribution:* Although real-world 6G environments are typically heterogeneous, IID distributions may still arise when AMFs exhibit

TABLE II
PERFORMANCE COMPARISON OF FIVE FEDERATION APPROACHES UNDER IID AND NON-IID CONDITIONS

Dirichlet α	Accuracy						MSE					
	Local	FedAvg	FedPer	FedProx	Ditto	HFWA	Local	FedAvg	FedPer	FedProx	Ditto	HFWA
0.1	63.20%	65.99%	66.85%	67.07%	68.65%	68.71%	0.0556	0.0490	0.0487	0.0477	0.0465	0.0461
0.2	63.47%	66.52%	67.29%	68.21%	69.08%	69.19%	0.0543	0.0489	0.0479	0.0474	0.0463	0.0459
0.3	63.67%	67.14%	68.16%	68.29%	69.15%	69.59%	0.0531	0.0485	0.0473	0.0470	0.0461	0.0458
0.4	64.21%	67.23%	68.34%	68.70%	69.27%	69.61%	0.0529	0.0474	0.0465	0.0463	0.0464	0.0451
0.5	64.59%	67.73%	68.38%	69.15%	69.31%	69.66%	0.0526	0.0468	0.0464	0.0465	0.0458	0.0449
0.6	64.69%	68.97%	68.79%	69.17%	69.52%	70.17%	0.0515	0.0456	0.0462	0.0458	0.0454	0.0445
0.7	64.83%	69.12%	68.94%	69.55%	69.67%	70.54%	0.0513	0.0454	0.0460	0.0453	0.0451	0.0433
0.8	65.04%	69.53%	68.97%	69.84%	70.02%	70.64%	0.0512	0.0450	0.0457	0.0442	0.0446	0.0429
0.9	65.26%	70.33%	69.25%	70.56%	70.85%	71.23%	0.0503	0.0447	0.0453	0.0433	0.0427	0.0425
1.0	65.48%	70.83%	70.27%	71.35%	71.44%	71.91%	0.0484	0.0438	0.0449	0.0423	0.0417	0.0411
IID	69.88%	72.42%	71.39%	72.25%	71.67%	72.25%	0.0455	0.0392	0.0403	0.0394	0.0393	0.0394

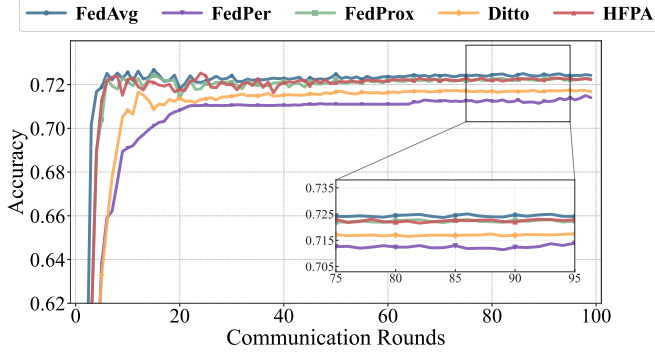


Fig. 8. Performance Comparison Between HFWA and Four Federated Approaches under IID Conditions.

similar user mobility patterns and traffic characteristics. Fig. 8 shows the performance under IID conditions, simulated by partitioning data across AMFs using a Dirichlet distribution with $\alpha = 1000$.

Specifically, all five FL algorithms gradually converge as the number of communication rounds increases, indicating that aggregating local knowledge enables effective model learning across AMFs. At the 90th communication round, HFWA achieves an accuracy of 72.25%, slightly lower than FedAvg by 0.15%, while outperforming FedPer and Ditto by 1.01% and 0.53%, respectively. This performance gap is mainly caused by the regularization term in HFWA, where the balancing factor μ constrains local updates to remain close to the previous global model. As a result, HFWA sacrifices slight performance under IID conditions to improve adaptability to heterogeneous data distributions in real 6G environments.

3) *Evaluation of HFWA under Various Non-IID Degrees:* Non-IID conditions represent heterogeneity in the data across clients. We simulate 100 AMF clients whose data distributions follow a Dirichlet process parameterized by α . We evaluate HFWA against four FL baselines in terms of accuracy and MSE, as shown in Table II.

Overall, HFWA consistently achieves the best performance across different alpha settings. When $\alpha = 0.1$, the Dirichlet distribution results in highly skewed data distributions across AMFs. In this case, the local and HFWA accuracy is 63.20% and 68.71%, respectively, indicating that FL facilitates effective data collaboration and sharing. When $\alpha = 1$,

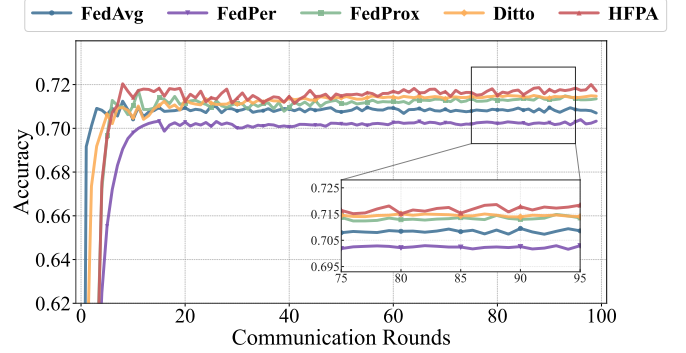


Fig. 9. Performance Comparison Between HFWA and Four Federated Approaches under Non-IID Conditions.

HFWA achieves an accuracy of 71.91%, surpassing FedAvg, FedPer, FedProx, and Ditto by margins of 1.08%, 1.64%, 0.56%, and 0.47%, respectively. To provide a more detailed illustration, we set $\alpha = 1$, as shown in Fig. 9. The superior performance of HFWA is attributed to its three key design components: RPO, LPE, and FWA. RPO aligns the global model W with region-specific data distributions, and LPE incorporates regional knowledge while preserving client-specific characteristics. FWA mitigates the aggregation bias introduced by heterogeneous data distributions and reduces the adverse effects of Non-IID heterogeneity.

4) *Ablation Study on Functional Modules of HFWA:* To evaluate the effectiveness of the three proposed modules in the HFWA framework, experiments are conducted with $\alpha = 1$. LPE, RPO, and FWA modules are combined in various configurations, and performance is evaluated using Accuracy, MSE, and MAE. The results are shown in Table III.

The ablation study shows that HFWA achieves the best performance when all three modules are integrated, reaching an accuracy of 71.91%, with MSE and MAE of 0.0411 and 0.0825, respectively. When each module is retained individually, performance consistently degrades, indicating that each component contributes to the framework's overall effectiveness.

Specifically, the LPE module improves the model's ability to capture localized feature representations. Its removal reduces prediction accuracy. Similarly, the RPO module enhances optimization robustness by refining parameter updates. Its absence

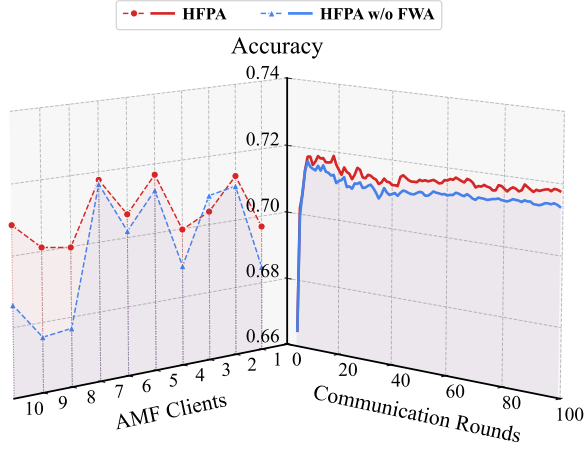


Fig. 10. Comparative Fairness Evaluation of HFGPA Across AMF Clients under FWA and without FWA Conditions

TABLE III
ABLATION STUDY OF HFGPA COMPONENTS: LPE, RPO, AND FWA

Methods			Accuracy	MSE	MAE
LPE	RPO	FWA			
			70.87%	0.0436	0.0886
✓			71.37%	0.0420	0.0855
	✓		71.20%	0.0426	0.0870
		✓	71.12%	0.0427	0.0873
✓	✓		71.65%	0.0414	0.0839
	✓	✓	71.41%	0.0419	0.0853
✓		✓	71.57%	0.0416	0.0842
✓	✓	✓	71.91%	0.0411	0.0825

leads to higher prediction errors. Finally, the FWA module dynamically adjusts aggregation weights to help mitigate data heterogeneity across clients. Without it, the model suffers from suboptimal aggregation and performance degradation.

5) *Evaluating the Fairness Performance of FWA in HFGPA*: The contribution of FWA within HFGPA is evaluated by covering both global convergence across AMFs and client-level performance. We evaluate the framework under the Dirichlet setting with $\alpha = 1$ in two configurations, HFGPA with FWA (HFGPA) and HFGPA without FWA (HFGPA w/o FWA). Fig. 10 presents the evaluation of the contribution of FWA within HFGPA along two complementary dimensions.

The experimental results indicate that FWA effectively enhances the stability and accuracy of HFGPA in Non-IID environments. Both HFGPA and HFGPA w/o FWA exhibit stable convergence as the number of communication rounds increases. At the 100th communication round, HFGPA reaches an accuracy of 71.75%, yielding an improvement of 0.57% over HFGPA w/o FWA. We further present the fairness performance across different AMF configurations in Fig. 10. These results indicate that FWA effectively mitigates cross-client data heterogeneity, thereby preventing model bias driven by clients with relatively small data volumes.

6) *Evaluating HFGPA under Local Differential Privacy Constraints*: To mitigate user privacy risks when uploading local model parameters, we incorporate local differential privacy

into HFGPA. As in previous studies [49]–[51], which evaluate privacy-preserving performance by monitoring client-specific changes in model accuracy, we apply the same evaluation metrics to HFGPA. In these experiments, we set $C = 5$ and $\delta = 10^{-5}$, and adjust ϵ to represent varying noise levels. Table IV presents the experimental results, and we analyze how changes in ϵ affect both privacy and model accuracy.

The experimental results indicate that the observed decrease in accuracy and the corresponding increase in MSE reflect the impact of the injected noise. These changes suggest that the added perturbations effectively disrupt the model updates. Specifically, as ϵ decreases, indicating increased noise injection, all FL algorithms show a consistent decrease in accuracy. Compared to other FL approaches, HFGPA exhibits a smaller drop in accuracy. In particular, when $\epsilon = 0.1$, which corresponds to the strongest level of noise perturbation, HFGPA achieves an accuracy of 39.89%. Nevertheless, it still outperforms all competing FL methods. As ϵ increases to 1, the accuracy of HFGPA improves to 71.41%, surpassing FedAvg and Ditto by 0.68% and 0.23%, respectively. Compared to the configuration without LDP, the model exhibits a slight performance decline. The observed reduction in performance aligns with previous findings that stronger privacy mechanisms generally lead to a slight degradation in model accuracy. This degradation is primarily caused by stochastic perturbations in local parameters introduced by the differential privacy mechanism, which disrupt the parameter update process. The resulting performance fluctuations stay within an acceptable range.

7) *Complexity Analysis*: In RelTime-PRE, an input sequence of length L is segmented into patches through Trajectory Patching with stride S , producing approximately L/S patches. The Transformer encoder operating on these patches incurs a computational complexity of $\mathcal{O}((L/S)^2)$. The subsequent BiLSTM module has a computational complexity of $\mathcal{O}(L(dH + H^2))$, where d denotes the input dimension and H the hidden state dimension. Consequently, the overall time complexity of RelTime-PRE can be approximated as $\mathcal{O}((L/S)^2) + \mathcal{O}(L(dH + H^2))$.

For the KD-Tree search, let N denote the total number of base stations. The algorithm maintains a max-heap of size K , and each heap update costs $\mathcal{O}(\log K)$. According to the worst-case analysis by Lee and Wong [52], the number of visited nodes in a k -dimensional KD-tree satisfies $V_k(N) = \mathcal{O}(kN^{1-1/k})$. In our paging scenario with $k = 2$, this simplifies to $V_2(N) = \mathcal{O}(\sqrt{N})$. Thus, the KD-Tree search time complexity is given by:

$$T(N) = \mathcal{O}(\sqrt{N} \log K). \quad (45)$$

Combining the model computation with the KD-Tree search, the overall time complexity of a single prediction is $\mathcal{O}((L/S)^2 + L(dH + H^2) + \sqrt{N} \log K)$.

8) *Comparative Analysis of Signaling Overhead for 6G-PFIP*: We further evaluate the signaling overhead of the proposed 6G-PFIP paging mechanism, building on the design above, which balances performance and privacy while tolerating a moderate loss in accuracy [53]. Therefore, we conduct a comparative analysis of the 6G-PFIP paging mechanism

TABLE IV
PERFORMANCE COMPARISON OF FIVE FEDERATION APPROACHES UNDER DIFFERENT NOISE SCALES

Noise Parameters			Accuracy					MSE				
ϵ	C	δ	FedAvg	FedPer	FedProx	Ditto	HFLPA	FedAvg	FedPer	FedProx	Ditto	HFLPA
0.1	5	1e-5	37.94%	36.44%	38.62%	39.61%	39.89%	0.2836	0.2917	0.2580	0.2371	0.2206
0.2	5	1e-5	63.74%	62.17%	64.19%	64.91%	65.21%	0.0510	0.0556	0.0492	0.0493	0.0489
0.3	5	1e-5	68.40%	68.21%	68.54%	68.92%	69.31%	0.0443	0.0455	0.0444	0.0442	0.0441
0.4	5	1e-5	69.43%	68.41%	69.84%	69.42%	69.63%	0.0434	0.0447	0.0434	0.0440	0.0439
0.5	5	1e-5	69.92%	68.95%	70.02%	70.02%	70.08%	0.0431	0.0443	0.0431	0.0434	0.0433
0.6	5	1e-5	70.29%	69.32%	70.37%	70.47%	70.51%	0.0429	0.0439	0.0430	0.0429	0.0428
0.7	5	1e-5	70.43%	69.60%	70.46%	70.62%	70.90%	0.0428	0.0439	0.0429	0.0428	0.0427
0.8	5	1e-5	70.53%	69.91%	70.66%	70.97%	71.01%	0.0427	0.0437	0.0428	0.0427	0.0427
0.9	5	1e-5	70.62%	69.98%	70.74%	71.16%	71.23%	0.0426	0.0437	0.0428	0.0426	0.0425
1.0	5	1e-5	70.73%	70.05%	71.01%	71.18%	71.41%	0.0426	0.0435	0.0427	0.0426	0.0419

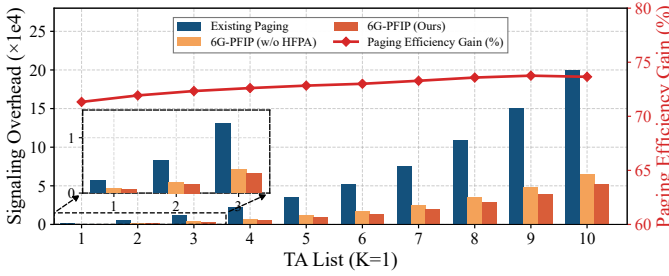


Fig. 11. Performance Comparison of Paging Signaling Overhead Between 6G-PFIP and Other Paging Schemes when $L_{pre} = 1$.

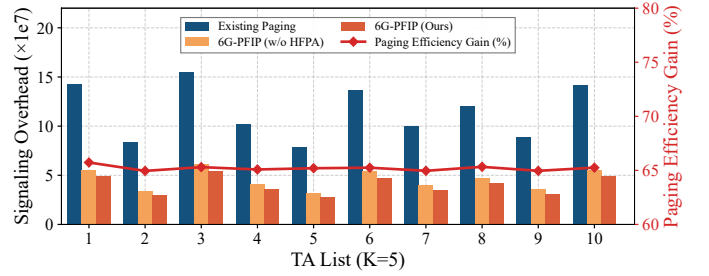


Fig. 12. Performance Comparison of Paging Signaling Overhead Between 6G-PFIP and Other Paging Schemes when $L_{pre} = 5$.

against both the existing paging mechanism and the 6G-PFIP without the HFLPA paging mechanism across various TA scenarios.

Fig. 11 and Fig. 12 illustrate the performance comparison of paging signaling overhead between 6G-PFIP and other paging schemes. To approximate realistic deployment, we configured 10 TA lists, each of which randomly contained 100 to 1,000 gNBs. Each gNB manages a user group of 10,000 to 20,000 UEs. Following 3GPP specifications, the paging signaling overhead of existing mechanisms is calculated as follows:

$$S_i^{\text{Existing}} = \lambda_i \cdot N_i, \quad (46)$$

Here, N_i denotes the number of gNBs in the i -th TA list, where $i = 1, \dots, 10$. Each paging event requires a full broadcast to all gNBs.

For the intelligent paging mechanism, the Paging Signaling Overhead is calculated as follows:

$$S_i^{\text{Intelligent}} = \lambda_i \cdot \mathbb{E}[S_i | \mathcal{C}_i], \quad (47)$$

where $\mathbb{E}[S_i | \mathcal{C}_i]$ denotes the expected signaling overhead for TA List i given the predicted candidate set $\mathcal{C}_i$.

Notably, we introduce a user behavior weight λ_i , which is calculated as follows:

$$\lambda_i = \lambda_{\text{total}} \cdot \frac{\sum_{j=1}^{N_i} w_{ij} U_{ij}}{\sum_{i=1}^M \sum_{j=1}^{N_i} w_{ij} U_{ij}}, \quad (48)$$

Where λ_i denotes the paging arrival rate of the i -th TA List, and λ_{total} represents the total system paging arrival rate. U_{ij} indicates the number of users managed by the j -th gNB in the i -th TA List, and w_{ij} represents the weight of this gNB,

reflecting its relative importance. M denotes the total number of TA Lists.

It is worth noting that the search scope within each TA list can be treated as a key variable in the intelligent paging process. Specifically, the parameter K denotes the number of candidate locations predicted by the model, which directly determines the number of target TA lists to be searched in the second-step paging. When $K = 1$, the prediction model outputs only a single predicted location, $L_{pre} = 1$. In this case, the paging procedure searches only one target TA list, resulting in a highly focused search region and minimal signaling overhead. However, the paging success rate may be sensitive to prediction errors. In contrast, when $K = 5$, the model outputs multiple candidate locations ($L_{pre} = 5$) and the paging procedure searches across multiple TA lists. This expands the search coverage and improves robustness against prediction uncertainty, but inevitably increases signaling overhead.

Fig. 11 presents the comparison of paging signaling overhead between 6G-PFIP and existing paging mechanisms across 10 TAs of varying scales when $L_{pre} = 1$. Specifically, in the TA1 region, which contains 120 gNBs, the conventional paging mechanism incurs a signaling overhead of 2,316. In contrast, the intelligent paging mechanism without HFLPA reduces the overhead to 814, while the proposed 6G-PFIP further decreases it to 664. These results demonstrate that 6G-PFIP improves user-carrying capacity by 71.33% compared with the conventional paging mechanism. Similar trends can also be observed as the TA scale increases. This improvement is due to the conventional paging scheme, which broadcasts paging messages to all TAs associated with the UE after the initial paging attempt fails. In contrast, 6G-PFIP leverages accurate

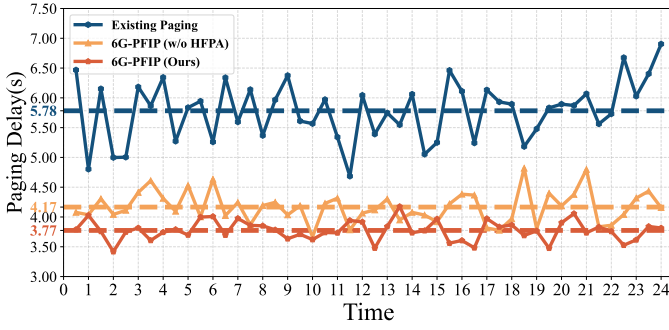


Fig. 13. Performance Comparison of 24-hour Paging Delay for a Single User Between 6G-PFIP and Existing Paging Scheme.

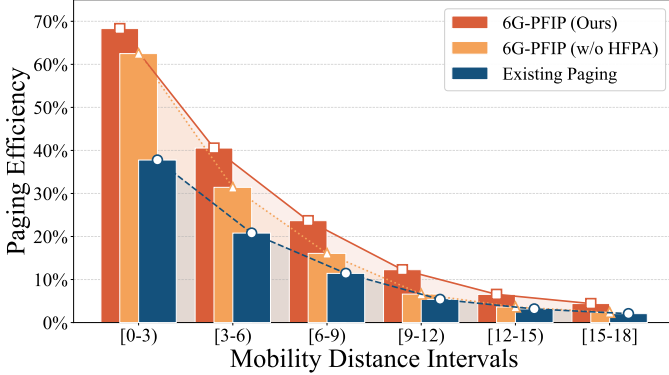


Fig. 14. Performance Comparison of Paging Efficiency Across Different Mobility Distance Intervals.

prediction of the target gNB to enable targeted secondary paging, thereby reducing unnecessary broadcast signaling.

Fig. 12 compares the paging signaling overhead of 6G-PFIP with existing paging mechanisms when $L_{\text{pre}} = 5$. Even as the user scale increases, 6G-PFIP consistently achieves stable improvements in paging efficiency. In the TA10 region, the signaling efficiency gain reaches 65.25%. Overall, paging efficiency decreases as prediction length increases due to greater uncertainty at longer horizons.

The reduction in signaling overhead in 6G-PFIP primarily results from replacing broadcast-based paging with prediction-driven, targeted paging. Unlike conventional mechanisms that broadcast paging messages across all TA lists, 6G-PFIP narrows the paging search space through UE mobility prediction, thereby reducing unnecessary signaling transmissions. In addition, the federated framework improves prediction accuracy and enables more precise selection of the target gNB. Regarding the impact of different values of K , smaller K values reduce signaling overhead but are more sensitive to prediction errors, whereas larger K values improve robustness at the cost of increased signaling overhead. This demonstrates a trade-off between paging efficiency and reliability.

VIII. PRACTICAL DEPLOYMENT

1) *Practical Deployment Evaluation of 6G-PFIP*: To evaluate the paging delay (PD) of 6G-PFIP, we built a physical testbed with two central servers and three physical hosts, forming a distributed service-function network with full core paging functionality. 3GPP TS 24.501 specifies T3513 as the

standard paging timer. This setting is supported in the Cisco Configuration Guide, which recommends a paging timeout of 1-10s. The PD is calculated as follows:

$$D^{(m)}(t) = T_0 + T_p \cdot \mathbb{E}[H^{(m)}(t)] + T_{fb} \cdot \mathbb{E}[F^{(m)}(t)], \quad (49)$$

where m denotes different paging schemes, T_0 captures the scheme-independent baseline latency, while T_p and T_{fb} characterize the delay contributions of selective paging and fallback paging, $\mathbb{E}[H^{(m)}(t)]$ and $\mathbb{E}[F^{(m)}(t)]$ reflects the average success probability of the selective paging and the fallback triggering probability, respectively.

Fig. 13 compares the paging delay over 24 hours for moving UE between 6G-PFIP and the existing paging scheme. The three dashed lines represent the average paging delay under the existing paging scheme, 6G-PFIP without HFPA, and the full 6G-PFIP, respectively. The complete 6G-PFIP achieves an average delay of 3.77 seconds, representing a 34.78% reduction compared to the existing scheme and a 9.6% improvement over the version without HFPA. This performance gain results from the combined effect of RelTime-PRE and HFPA in 6G-PFIP. Overall, 6G-PFIP consistently maintains low paging latency under various scenarios, showing strong potential for large-scale deployment.

2) *Evaluation of Paging Efficiency Across Different User Mobility Distance Intervals*: To evaluate paging efficiency under high mobility, we designed a simulation scenario comprising five TAs to compare the traditional scheme against 6G-PFIP. Each TA is equipped with 20 gNBs, covering a range of approximately 3 km.

As illustrated in Fig. 14, 6G-PFIP consistently outperforms the traditional scheme and 6G-PFIP without HFPA. Specifically, in the 0–3 km interval, 6G-PFIP achieves a paging efficiency of 68.31%, representing improvements of 80.9% over the existing paging scheme and 9.3% over 6G-PFIP (w/o HFPA). This is primarily attributed to its precise location prediction, which facilitates prompt identification of candidate gNBs following a last-gNB failure. In the 3–12 km medium-mobility interval, the scheme exhibits a significant advantage, validating its capability to capture cross-TA mobility characteristics effectively. Even in the 12–18 km high-mobility interval, the efficiency of 6G-PFIP remains significantly higher than that of the traditional scheme, demonstrating that it maintains superiority in extreme scenarios.

IX. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper proposed a Privacy-Preserving and Personalized Federated Intelligent Paging Mechanism called 6G-PFIP. Specifically, 6G-PFIP introduced an intelligent decision model, RelTime-PRE, that combines a Transformer and a BiLSTM to capture both global trends and local temporal patterns for intelligent paging prediction. Furthermore, a Personalized Hierarchical Federated Paging Architecture with Fairness (HFPA) was designed to mitigate the impact of data heterogeneity on model performance by optimizing regional and local personalization and performing fair weighted aggregation. In addition, HFPA enhances privacy protection by

incorporating local differential privacy into model parameter updates. Extensive evaluations on a real-world Shanghai Telecom dataset demonstrated that 6G-PFIP significantly reduces paging latency and signaling overhead compared with existing paging mechanisms. Rigorous analysis further validated that 6G-PFIP provides privacy preservation and personalization capabilities over existing paging mechanisms.

B. Future Work

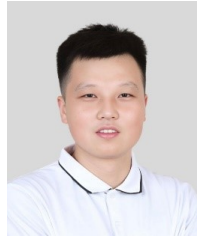
Although 6G-PFIP improves paging efficiency, adaptability, and privacy preservation, further improvements in prediction accuracy, system robustness, and practical deployment are still required. In future work, we will extend RelTime-PRE by incorporating spatiotemporal features and gNB topology through graph neural networks, while refining HPPA with meta-learning to support rapid adaptation in personalized federated environments. In addition, we will further enhance system robustness through adaptive strategies and lightweight online learning mechanisms. From a system perspective, we will investigate the deployment of 6G-PFIP in large-scale dynamic environments with heterogeneous user distributions and time-varying traffic, while reducing the communication overhead and latency introduced by federated training and privacy-preserving mechanisms. We also plan to validate the proposed framework on large-scale real-world datasets and network testbeds to further improve practical applicability.

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